



Ph.D. Handbook 2026

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Message from the Chair and Director of Graduate Studies (DGS)

Professor Amir Ali Ahmadi



Professor Ludovic Tangpi



Welcome to the Department of Operations Research & Financial Engineering (ORFE). We wish you tremendous academic success over the next few years. At Princeton, you will be among the best, brightest and most creative talents of your generation. ORFE is a small, yet very interdisciplinary department. We trust that here you will thrive as you deepen and widen your academic interests and become experts in your respective fields of research.

ORFE is also a very friendly department where Undergraduate and Graduate students, as well as Faculty and Staff interact in a supportive atmosphere. We hope that you will easily integrate into the community. Please do introduce yourself to fellow students and to faculty, and do not hesitate to reach out to us if you have any questions or concerns. We look forward to engaging with you over the next few years.

Go Tigers!

1. Ph.D. Program Requirements

In their first year, students take the six core courses in statistics, probability and optimization. By the end of the second year, at least two advanced courses are passed, and two semesters of directed research are completed under the direction of a faculty adviser in the student's area of interest in preparation for the general examination (Generals). The general examination is taken in the spring of the second year of study. Usually, beyond the general examination, two to three years are needed for the completion of a suitable dissertation. Upon completion of the thesis and acceptance of the dissertation by the department and the Graduate School, the candidate is admitted to the final public oral examination (FPO).

In the first year of graduate study, students must take and pass all six core courses with a letter grade of B- or better. Please note that core courses cannot be dropped.

Core courses:

ORF 522	Linear & Nonlinear Optimization (Fall)
ORF 523	Convex and Conic Optimization (Spring)
ORF 524	Statistical Theory and Methods (Fall)
ORF 525	Statistical Foundations of Data Science (Spring)
ORF 526	Probability Theory (Fall)
ORF 527	Stochastic Calculus (Spring)

Qualifying exams: Qualifying exams are held in September of the second year. Students can place out of the qualifying exams by receiving at least an A- grade in four of the core courses. If a student received fewer than 2 A- or better grades in the 6 first-year courses, they cannot take the qualifying exams, and their Ph.D. candidacy ends. See the Ph.D. Qualifying and General Examinations for details.

ORF 509 & 510: Before taking the general exam, Ph.D. students must complete two research projects under the supervision (or co-supervision) of an ORFE faculty member. Students are required to submit a written report on both projects. The results of ORF 510 are presented in the oral part of the general exam. ORF 509 must be passed with a B+ or better. ORF 510 must be passed with a B+ or better after vote of the faculty.

Advanced courses: Before taking the general exam, Ph.D. students must take two additional approved advanced courses at the 500 level, which must be passed with at least a B+ grade. These two advanced courses can be chosen from the following list:

- ORF 504 Financial Econometrics
- ORF 505 Statistical Analysis of Financial Data
- ORF 531 Computational Finance in C++
- ORF 535 Financial Risk and Wealth Management
- ORF 538 PDE Methods in Financial Mathematics
- ORF 542 Stochastic Control and Stochastic Differential Games
- ORF 543 Deep Learning Theory
- ORF 544 Stochastic Optimization
- ORF 550 Special Topics in Probability: Probability in High Dimension
- ORF 569 Special Topics in Statistics, Operations Research and Financial Engineering (e.g., *Theory and Practice of Deep Learning, Probabilistic Theory of Network Games and Mean Field Interactions, Topics in Game Theory*)

Courses of potential interest in other departments and programs:

- APC 503 Analytical Techniques in Differential Equations
- COS 511 Theoretical Machine Learning
- COS 521 Advanced Algorithm Design
- COS 528 Data Structures and Graph Algorithms
- COS 534 Fairness in Machine Learning
- ECO 525 Asset Pricing
- ECO 526 Corporate Finance
- ECE 535 Machine Learning and Pattern Recognition
- ECE 538B ST: Information Sciences and Systems Large Scale Optimization for Data Science
- MAE 546 Optimal Control and Estimation
- MAT 522/APC 522 Introduction to PDE
- MAT 527 Topics in Differential Equations: Global Solutions of Nonlinear Evolutions
- MAT 572/APC 572 ST: Introduction to Combinatorial Optimization
- PHY 521/MAT 597 Introduction to Mathematical Physics

General Exam (at the end of the second year): This is an oral exam including a presentation of the student's research and research plan.

Teaching Requirement: The department has a teaching requirement of at least a full Teaching Assistant (TA) assignment or two half TA assignments potentially beginning in the fall of the student's 2nd year depending on the department's teaching needs. Requirements for teaching include:

- Passing the English Language Proficiency Exam by August after the first year of study (if applicable).
- Attending the mandatory TA Orientation in fall of the 2nd year of study.

RCR class: All Ph.D. candidates are also required to take EGR501 before the end of their second year. This is a half-semester course that will educate graduate students in engineering in the responsible conduct of research. The lectures provide theoretical background information as well as case studies about ethics in day-to-day research situations, publishing and peer-reviewing,

student-adviser relationships, collaborative research, as well as in the big picture and considerations of long-term impact. The course is graded on a pass/fail basis.

FPO: Research dissertation and Final Public Oral Exam (FPO).

Research & Thesis Adviser

The core courses establish a solid foundation, giving tools required for research in the disciplines covered by the department. The funding provided by the university for the first year provides freedom and time for students to acclimate to the new environment of graduate school, and to make a wise choice of research direction and adviser. Some may know early on the direction they wish to pursue, but for many, it is worth exploring a number of courses taught by a variety of professors before deciding.

Students ordinarily match with an adviser by the end of the first year of study to begin research in preparation of the general exam. Normally a student may not enter the second year without a research adviser of record.

The directed courses, ORF 509 and 510 are taken with a faculty adviser, with a grade received for both by the time of the general exams. Students typically take these courses in the fall and spring of the second year.

The department is small and there are plenty of opportunities to meet and talk with faculty about research interests. Please be pro-active in doing so as you discover your areas of interest. At various times, some faculty might be too busy to take on new students: do not take it personally if a faculty member is unable to take you on as an advisee, there are plenty of excellent research opportunities available to you. If a student works with a faculty adviser who is not a core member of the ORFE faculty, there must also be a co-adviser who is an ORFE core faculty member, and who is seriously involved in co-advising the student. All external advising arrangements must be approved by the DGS.

Ph.D. Qualifying and General Examinations

The requirements for the Ph.D. degree include passing the qualifying exam requirements, the general examination, submission of an acceptable dissertation, and passing the final oral examination (thesis defense).

Qualifying Examination Procedure

Each student must satisfy qualifying requirements. Qualifying exams are offered in September of the student's second year. A student who obtained a grade of A- or better in 4 of the six required core courses will be exempt from taking the qualifying exams. If this is not the case and the student obtained a grade of A- or better in at least 2 of the six required courses, the student will meet with the DGS to decide which exams must be taken to satisfy the requirements. The optimization exams are based on ORF 522 and ORF 523. The statistics exams are based on ORF 524 and ORF 525. The probability exams are based on ORF 526 and ORF 527. Students who

received fewer than 2 A- or better grades in the 6 first-year courses are not eligible to take qualifying exams, and their Ph.D. candidacy ends.

The qualifying exams shall not exceed 90 minutes per course and are administered by 2 ORFE faculty. The examining faculty will decide if the qualifying exam will be written or oral, in consultation with the DGS, and will notify the student ahead of time. If the qualifying exam is

The results of the qualifying exam(s) are determined by a vote of the faculty.

Requirements for passing Ph.D. qualifying examination

- at least 4 A- grades in the 6 core first-year courses: pass (no qualifying exams)
- 3 A- in the 6 core first-year courses, and passes one qualifying exam: pass
- 2 A- in the 6 core first-year courses, and passes two qualifying exams: pass

General Examination Procedure

ORFE students take the general exam in April or May of their second year. By the end of the Spring semester, students should have met the qualifying examination requirements, have taken and passed ORF 509 and ORF 510, and have passed with a B+ or higher grades two approved

The general exam consists of two parts, a written and an oral part, both covering the student's primary area of specialization. The written part requires taking and passing, with a B+ or higher grade, two approved advanced courses at the graduate (500) level beyond the 6 core first-year courses. The student's adviser and the DGS must approve these two courses.

For each student, the student and adviser select an examination committee. The DGS will approve the committee. Normally, the committee members comprise the student's adviser(s) and two additional ORFE faculty or associated faculty. The committee will administer the oral exam, evaluate the student's performance in research and overall knowledge of their field, and make a recommendation to the department faculty. A departmental faculty vote determines the outcome. The oral exam may be up to 3 hours in length.

Information on the oral exam: Before the exam, the student is required to submit a comprehensive written report on the research conducted in ORF 509-510. It is due one week before the exam takes place. The report serves as the basis for the student's presentation. The purpose of the presentation is to explain the research the student has done so far and plans to do in the future. Examining faculty may ask questions on the presentation and on any other material deemed appropriate for a comprehensive examination.

The possible outcomes of the general exam are as follows:

Student passes the exam and is admitted to Ph.D. candidacy.

Student fails the exam but is given the option to retake the exam at a future date.

Student fails the exam, and their Ph.D. candidacy ends.

Summary of first and second year requirements

- First year: By the end of the first year, students must take and pass all six core courses with a letter grade B- or better; and pass their qualifying exams (see above for qualifying

exams requirements). Moreover, students must [fulfill the Teaching Assistant requirements](#).

- Second year: By the end of the second year, students must take and pass ORF 509 & 510 as well as two additional approved advanced classes at the 500 level with a letter grade of B+ or better. Students are also required to take the RCR course and pass the General Examination.

Qualifying for the Master's degrees

Master of Arts (MA.).

The Master of Arts (M.A.) degree is normally an incidental degree on the way to full Ph.D. candidacy and is earned after a student successfully passes the general examination.

It may also be awarded to students who, for various reasons, leave the Ph.D. program, provided that these requirements have been met. To be considered for the MA degree, students must fulfill all the first and second year Ph.D. requirements except for passing ORF 510 and the General Exam. (ORF 522, 523, 524, 525, 526, 527, Two advanced courses, ORF 509)

Master of Science in Engineering (MSE).

To be considered for the MSE degree, students must fulfill all the first and second year Ph.D. requirements, submit a Masters thesis and pass a Final Public Oral examination (FPO). Ph.D. students who have been awarded the incidental M.A. are not eligible to earn an M.S.E.

Both the MA and MSE degrees are awarded by vote of the faculty and approval of the Graduate School. Students cannot be considered for both degrees.

1.5 Annual Reenrollment

Second Year Reenrollment

Upon passing the general examination, the recommendation for readmission for the next academic year is made by the student's thesis adviser and approved by the department faculty.

Research Advising Committee for Ph.D. Candidates

If the thesis adviser plans to recommend against the readmission of the student, an examination committee must be formed to determine the progress of the student's work and the recommendation must be made by the committee and approved by the department faculty.

Third- and Fourth-Year Advisory Committee

If *requested* by either the student's thesis adviser, or the student, or the Director of Graduate Studies, an examination committee shall be appointed by the department Chair to examine the progress of the student. The committee would be composed of the student's adviser in addition to two other ORFE faculty members.

The student will present a seminar to this committee describing the proposed area of research, the background work completed in preparing for the research effort, the intended avenues of

1.6 Dissertation and Post-Generals Courses

Completion of a suitable dissertation usually takes two to three years beyond the general examination. During those years the student is encouraged to select an area outside his/her specialty and to broaden his/her base of knowledge by enrolling in courses in that area. It is also expected that additional courses will be taken related to the student's specialty area. Upon completion and acceptance of the dissertation by the department, the candidate will be admitted to the final public oral (FPO) examination.

1.7 Final Public Oral (FPO): Department Instructions

The readers and FPO examiners must be approved first by the ORFE department chair, and then by the Graduate School. The first reader is the principal adviser, and the second reader is an ORFE faculty member. An FPO examination committee and readers approval request must be submitted to the Department Chair's office. Please [submit the request at least two months before the anticipated FPO date to Kimberly Lupinacci in the Graduate Administrator's office](#). If any reader is not a Princeton ORFE professor, the request should include a written justification for why an external reader is needed. The committee of examiners for the FPO must include no fewer than two current ORFE faculty members, and no fewer than three current Princeton faculty members. If a committee member is not an ORFE faculty member, the request must include written justification for why an external committee member is needed.

When the committee and readers are approved by the department, the student completes the advanced degree application via Tigerhub. Once approved, the student can send the thesis to the readers. After the readers submit their completed reports, the Graduate School must approve the requested FPO date and time.

Please note that FPO's are not approved for the months of June, July and August

2 Other Regulations

Satisfactory Academic Progress

To establish the foundation for satisfactory academic progress, the Graduate School and academic departments expect the following from enrolled students, as evidence of their successful engagement with graduate work and for them to continue to receive their stipends and other benefits of enrollment:

- Full-time Commitment. Graduate study at Princeton, at both the doctoral and master's levels, requires full-time commitment to study and research on the part of students.
- Presence: Students must be visibly present in the department and on campus, unless *In Absentia* or on an approved Leave of Absence. As importantly, students must be intellectually “present,” that is, noticeably engaged in the normal work of their degree program—course-taking, paper-writing, research, teaching, seminar attendance.
- Attendance: Attendance at the ORFE Colloquium is expected of all graduate students on a weekly basis. Students are also encouraged to attend at least one additional seminar each week that is broadly relevant to their research interests
- Production: Students must be producing work of good quality, at the appropriate and expected stages of their degree program and showing the products of their study and research to the faculty for evaluation.
- Communication: Students must regularly communicate with and respond in a timely manner to communications from their department graduate program administrator, director of graduate studies, adviser(s), committee members, and other faculty members as appropriate. This is a reciprocal responsibility, and graduate students should, therefore, expect regular and timely communication to maintain satisfactory academic progress.

Failure to perform according to these guidelines may result in, for example, deferred reenrollment or denial of reenrollment during the spring term review or termination at mid-year.

Academic Regulations and Fraud

The Graduate School has become increasingly concerned about academic fraud. Please take time to read information on “General Requirements for the Acknowledgment of Sources in Academic Work” and “Definitions of Academic Violations under the Jurisdiction of the Faculty-Student Committee on Discipline” contained in “Rights, Rules and Responsibilities”. Upon entering Princeton, students receive information concerning this important subject and University requirements on acknowledgment of sources and academic fraud.

2.3 Changes in Course Status

Any changes (grading options, dropping or adding courses etc.) that a student may need to make to their registered courses should be discussed with their adviser or DGS prior to making changes. A student may change their status in a course no later than two weeks prior to the last day of the regularly scheduled class of the semester. Core course changes must be approved by DGS in writing.

2.4 Auditing Courses

1. Courses selected by pre-General students must be taken for a letter grade and cannot be audited.
 2. Post-General students may sign up for courses as audit.
- Post-General students may take courses on a pass/fail (P/F) basis with the approval of their adviser and the course instructor.

Generally, courses taken on a P/F basis should be outside of the student's primary area of interest.

2.5 Teaching Assistants (TAs)

Normally, graduate students receive appointments as teaching assistants (TAs) beginning their second year. All students are required to attend TA training prior to term. Before an international student can be appointed as a TA, they need to pass the English Language Program (ELP) test administered by the Graduate School, and to complete any required ELP courses.

2.6 Part-time Employment

For Information regarding Part-time Employment, please refer to the graduate school website on [Employment](#) or the Davis International [Employment](#) website.

2.7 Leave of Absence

All ORFE Graduate Students are required to be on campus during term time. Any time away from campus exceeding 3 business days must be approved by the student's adviser, or by the DGS for 1st year students without an adviser. Students are responsible for [submitting a request](#) at least 2 weeks prior to leave and is not approved until the student receives a decision from the faculty member with notification to the Graduate Program Administrator.

2.8 Holiday, Vacation, and Travel

All time away from campus during the semester (except weekends and university holidays) must be approved in writing by the adviser and, if a Teaching Assistant, by the DGS and course instructor. An email requesting time away from campus with leave and return dates and reason for absence is appropriate. In the case of first year students that may not yet have an adviser, you must submit your request via email through Kim Lupinacci in the Graduate Office who will obtain approvals from the Director of Graduate Studies.

Graduate study is understood to be a full-time commitment on the part of students. During an academic year, which includes the summer, graduate student degree candidates may take up to (but no more than) four weeks of vacation, including any days taken during regular university holidays and scheduled recesses (e.g., the fall- and spring- term breaks and inter-term break). The specific periods taken as vacation must not conflict with the student's academic responsibilities, coursework, research, or teaching, and should be discussed in advance with the director of graduate studies or adviser.

If a student receives financial support for graduate study for only part of the year (e.g., regular term time, August 1 to May 31), then the amount of vacation should be prorated accordingly. If a student receives summer support and has taken the allowed vacation during regular term time, August 1 to May 31, then they should not take additional vacation time during the summer months of June and July.

If a student holds an external fellowship whose terms may conflict with this guideline, the student should consult first with the director of graduate studies or adviser.

IMPORTANT NOTICE: All ORFE Graduate Students are required to be on campus during term time. Any time away from campus exceeding 3 business days **must be approved** by your adviser, or by the DGS for 1st year students without an adviser. You are responsible for [submitting a request](#) at least 2 weeks prior to leave.

Please visit the [Registrar](#) for the Princeton University Academic Calendar.

2.9 Travel Support Request Process

Please review all [ORFE Travel and Expense requirements](#) prior to booking supported travel. All travel funding support opportunities are to be requested in the following order: [Adviser](#), [SEAS](#), [Graduate School](#), Department. Required department forms are located on the ORFE website, [Graduate Forms](#).

Miscellaneous Information

- Watch your emails for the following departmental events. Your presence at these events is appreciated.
 - Graduate Welcome Reception (September)
 - ORFE Socials (monthly)
 - Academic job panel (early Fall semester)
 - Department Holiday Party (December)
 - Seminars: Statistics (Mondays), ORFE colloquium (Tuesdays), Probability (alternate Wednesdays), Financial Mathematics (alternate Wednesdays), Optimization (Thursdays).
- klupinac@princeton.edu Graduate Student Affairs Office is 120 - Sherrerd Hall, extension 8-4018,
- E-mail is how department correspondence is transmitted. PLEASE check it regularly.
- <http://www.princeton.edu/tigerhub> Payroll checks are put in your mailbox on the last working day of the month. Go Green! Sign up for direct deposit via [TigerHub](#).
- Mailboxes are located in 221 - Sherrerd Hall. **Please check your mailbox.** The door to the mailroom is to be kept locked at all times.
- The address for package delivery is your name, Princeton University, 98 Charlton Street, Sherrerd Hall, Princeton, NJ 08544
- You may use the copier as a scanner in 108 - Sherrerd Hall.
- Graduate Student Kitchen is available to both graduate students and faculty in the ORFE department. Please keep this space clean.
- Graduate Lounge, the open area located on the second floor provides a meeting space for graduate students and faculty. This is a common area shared by all. This space is not to be used for office hours.

Important Contacts

Chair, Professor Mete Soner, 609-258-5130

Director of Graduate Studies, Professor Ludovic Tangpi, 609-258-4558

Department Manager, Connie Brown, 609-258-5422

Graduate Program Administrator, Kimberly Lupinacci, 609-258-4018

McCosh Health Center Emergency, 609-258-3139

Public Safety Non-Emergency 609-258-1000 Emergency 9-1-1

Welcome to the department, we wish you success as you pursue your graduate degree!